

A PROJECT TITLE
ON
CUSTOMER CREDIT SCORING USING XGBOOST AND LOGISTIC REGRESSION

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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By

Name : UPPALA SATHVIK
Name : NARAYANAGARI
PRUDVIRAJ GOUD
Name : P. NAKSHATRA

roll : 2520030226
roll : 2520030227
roll : 2520030243

Under the Esteemed guidance of

GADDAM RAVINDRA BABU

CSE

Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF Computer Science and Engineering

Name of Title: CUSTOMER CREDIT SCORING USING XGBOOST AND LOGISTIC REGRESSION

1. Abstract:

Customer Credit Scoring Using XGBoost and Logistic Regression is a machine learning-based system designed to evaluate the creditworthiness of customers and assist financial institutions in making reliable lending decisions. Credit scoring plays an important role in banking and financial services because it helps determine the likelihood that a customer will repay a loan on time. Traditional credit assessment methods can be time-consuming and may depend heavily on manual evaluation. This project aims to develop an automated and data-driven approach for classifying customers based on their credit risk.

The proposed system uses customer-related financial and demographic attributes such as income, age, credit history, loan information, repayment behavior, outstanding debts, and other relevant factors to predict whether a customer represents a low-risk or high-risk credit profile. Before model development, the dataset is cleaned and preprocessed by handling missing values, removing inconsistencies, encoding categorical variables, and selecting relevant features. Appropriate data preprocessing techniques are applied to improve the quality and reliability of the predictions.

Two machine learning algorithms, Logistic Regression and XGBoost, are implemented and compared. Logistic Regression provides a simple and interpretable classification approach, while XGBoost is an advanced ensemble learning algorithm capable of capturing complex relationships and patterns within the data. The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to determine their effectiveness in identifying credit risk.

The comparison helps identify which model provides better predictive performance while maintaining reliability and interpretability. The proposed system can support financial institutions in reducing credit risk, improving loan approval decisions, minimizing potential defaults, and providing faster customer assessments. Overall, the project demonstrates how machine learning can be effectively applied to customer credit scoring and contribute to more efficient, consistent, and data-driven financial decision-making.

2. INTRODUCTION:

Customer Credit Scoring Using XGBoost and Logistic Regression is a machine learning project developed to predict the creditworthiness of customers based on their financial and personal information. It helps banks and financial institutions identify high-risk and low-risk customers and reduce the chances of loan defaults. The project uses Logistic Regression and XGBoost to analyze customer data such as income, age, credit history, loan amount, and repayment behavior. The performance of both models is compared using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The better-performing model can then be used to support faster and more reliable credit decisions.

3. SOFTWARE REQUIREMENTS:

The project requires **Windows 10/11, Linux, or macOS** as the operating system. **Python 3.x** is used as the main programming language. The project can be developed using **Jupyter Notebook, GoogleColab, or Visual Studio Code**. The main libraries required are **Pandas and NumPy** for data processing, **Scikit-learn** for Logistic Regression, preprocessing, and model evaluation, and **XGBoost** for building the XGBoost classification model. **Matplotlib and Seaborn** are used for creating graphs and visualizing the data and model results. A web browser such as **Google Chrome, Microsoft Edge, or Firefox** is required for accessing Jupyter Notebook or Google Colab.

Signature of the Guide

Course Instructor